

Milestone 1 Report

Smart MCQ Solver Challenge

Course: BSDA2001P – Introduction to Deep Learning and Generative AI

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Objective

Milestone 1 helps us to understand and apply the foundation to a strong Natural Language Processing (NLP) baseline for solving the multiple choice questions. The work is most focused on understanding text processing, process of exploratory Data Analysis (EDA), vector space representations, semantic similarity using TF-IDF, and the Mean average precision (MAP@3) evaluation metric which are required specifically for NLP pipelines.

Dataset Description

The dataset consists of multiple choice questions where each question contains a prompt along with five answer choices labeled A,B,C,D and E.

And the goal of the student is to predict the top three most probable answers for every question.

For the data there were 3 files given

1. *train.csv* contains the training questions along with the correct answer labels.
2. *test.csv* contains the test questions without answers, students need to generate predictions for this file.
3. *Sample_submission.csv* demonstrates the correct submission format for the competition.

The columns of the files contain —

- *id* unique identifier for each question.
 - prompt is the question or problem statement.
 - *A, B,...,E* five possible answer choices.
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Exploratory Data Analysis (EDA)

The training dataset was inspected to understand:

- Dataset dimensions
- Missing values
- Answer distribution
- Vocabulary statistics

This analysis ensured that the dataset was suitable for subsequent NLP preprocessing.

Text Preprocessing

The following preprocessing steps were performed:

- Converted all prompts to lowercase.
- Removed punctuation using Python's `string.punctuation`.
- Tokenized text using white space splitting.
- Preserved the original dataset by performing preprocessing on a copied dataframe.

Removal of Stop Word

Standard English stop words from:

```
sklearn.feature_extraction.text.ENGLISH_STOP_WORDS
```

were used to analyze token reduction for individual prompts.

Vocabulary Analysis

Computed the following

- Frequency distribution of answer labels.
- Vocabulary size of cleaned prompts.
- TF-IDF vocabulary size after combining prompts and answer options.

TF-IDF Feature Extraction

Why TF-IDF is chosen

TF-IDF is selected as the first baseline because it gives a higher importance to critical words while reducing the influence of frequently occurring terms.

It is a simple representation of textual data before moving to more advanced embedding technique.

A baseline vector-space model was built using:

- `TfidfVectorizer(stop_words="english")`

The vectorizer was fitted on the combined text consisting of:

- Prompt
- Option A
- Option B
- Option C
- Option D
- Option E

This generated the vocabulary used throughout the similarity calculations.

Cosine Similarity

Cosine similarity was calculated between:

- Prompt vs Option A
- Prompt vs Option B
- Prompt vs Option C
- Prompt vs Option D
- Prompt vs Option E

The option with the highest similarity score was considered the predicted answer.

MAP@3 Evaluation

Implemented the MAP@3 evaluation metric used in the competition.

For every question:

1. Rank options according to cosine similarity.
2. Select the top three predictions.
3. Calculate MAP@3.
4. Compute the average MAP@3 across the complete training dataset.

Majority Class Baseline

A static baseline was created by always predicting the three most frequent answer labels from the training set.

This provides a simple benchmark for evaluating more advanced models.

TF-IDF Similarity Baseline

A simple retrieval-based baseline was implemented using:

- TF-IDF
- Cosine Similarity

Pipeline:

Prompt → TF-IDF Vectorization → Cosine Similarity → Rank Options
→ Top-3 Prediction

This serves as the first intelligent NLP baseline for the project.

Observations

- Text preprocessing significantly reduces vocabulary noise.
 - TF-IDF provides an interpretable baseline for semantic matching.
 - Lexical similarity alone is insufficient for solving reasoning-intensive MCQs.
 - The TF-IDF baseline demonstrates the limitations of bag-of-words representations for complex question answering tasks.
 - More advanced embedding methods and transformer-based architectures are expected to improve performance in later milestones.
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Libraries Used

- Pandas, Numpy, Scikit-learn, TF-IDF Vectorizer, Cosine Similarity

Results

	Metric	Value
Training Samples		2000
Prompt Vocabulary Size		859
TF-IDF Vocabulary Size		2762
Majority Baseline MAP@3		0.42125
TF-IDF Baseline MAP@3		0.29617